

BIOS8005: SELECTED TOPICS IN BIOSTATISTICS

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

Selected Topics in Biostatistics

Subject Code

BIOS - Biostatistics

Course Number

8005

Academic Unit

Biostatistics (BIOS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

R8 - Research Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

The course aims to introduce research students to one active, advanced and specialized field in Biostatistics (e.g. causal inference, genomic analysis, high-dimensional data analysis, post-model selection, reinforcement learning, survival

analysis, time series analysis, regularised regression, generalized linear models, regression for counts, etc.). It will help students to develop a solid and comprehensive understanding of the fundamental concepts, methods and theories in the chosen field and equip them with necessary techniques and knowledge for conducting independent and innovative research.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Explain the fundamental concepts and methods.	20	x	x	
2	Develop a solid understanding of the techniques.	20	x	x	
3	Conduct a thorough reading of the literature and know current state-of-the-art tools.	20	x	x	x
4	Apply the techniques and methods to real data applications.	20	x	x	x
5	Develop independent research skills and abilities	10	x	x	x
6	Effectively communicate and present research results	10	x	x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lectures	Learning through teaching is primarily based on lectures	1, 2, 3, 4, 5, 6	3 hours/week
2	Assignments	Learning though take-home assignments helps students understand the key concepts and acquire the techniques	1, 2, 3, 4, 5, 6	After class

Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Assignments	1, 2, 3, 4	30	Help to train students with basic knowledge, concepts, and analysis techniques	Yes
2	Midterm/quizzes	1, 2, 3, 4, 5, 6	20	Test students' capabilities in applying the knowledge to solve relevant problems	No

Continuous Assessment (%)

50

Examination (%)

50

Examination Duration (Hours)

3

Minimum Continuous Assessment Passing Requirement (%)

40

Minimum Examination Passing Requirement (%)

40

Additional Information for ATs

Examination questions are designed to see how well students have achieved the learning objectives and acquired the requisite techniques for problem-solving.

To pass the course, students are required to obtain a minimum of 40% in continuous assessment and a minimum of 40% in the examination.

Assessment Rubrics (AR)**Assessment Task**

Assignments (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Problem solving skills

Excellent

(A+, A, A-) Consistently exhibits adept comprehension of biostatistics concepts and methods and their practical implementation

Good

(B+, B, B-) Sufficient knowledge in applying biostatistics concepts and methods to moderately challenging problems

Fair

(C+, C, C-) Displays basic ability to use biostatistics concepts and methods and their applications to routine problems

Marginal

(D) Understand basic biostatistical issues related to routine problems

Failure

(F) Inadequate comprehension of biostatistics concepts and methods and lacks the ability to apply them to problem-solving

Assessment Task

Midterm/quizzes (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Problem solving based on comprehensive understanding

Excellent

(A+, A, A-) Exhibits a thorough grasp of time to biostatistics concepts and methods and effectively applies them to intricate problems

Good

(B+, B, B-) Displays sufficient understanding of biostatistics concepts and methods and effectively applies them to moderately challenging problems

Fair

(C+, C, C-) Shows basic comprehension of biostatistics concepts and methods and applies them to routine problems

Marginal

(D) Displays basic understanding of biostatistical concepts and their relationships to routine problems

Failure

(F) Insufficient grasp of biostatistics concepts and methods and lacks the ability to apply them to problem-solving

Assessment Task

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Problem solving based on comprehensive understanding

Excellent

(A+, A, A-) Consistently exhibits a deep understanding of biostatistics concepts and methods and effectively applies them to complex problems

Good

(B+, B, B-) Effectively applies biostatistics concepts and methods to moderately complex problems, demonstrating sufficient understanding

Fair

(C+, C, C-) Applies biostatistics concepts and methods to simple problems with a basic understanding

Marginal

(D) Displays basic understanding of biostatistical concepts and their relationships to routine problems

Failure

(F) Lacks understanding of biostatistics concepts and methods and cannot apply them to problem-solving

Assessment Task

Assignments (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Problem solving skills

Excellent

(A+, A, A-) Consistently exhibits adept comprehension of biostatistics concepts and methods and their practical implementation

Good

(B+, B) Sufficiently applies biostatistics concepts and methods to moderately challenging problems

Marginal

(B-, C+, C) Displays basic grasp of biostatistics concepts and methods and their application to standard problems

Failure

(F) Insufficient comprehension of biostatistics concepts and methods and lacks the ability to apply them to problem-solving

Assessment Task

Midterm/quizzes (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Problem solving based on comprehensive understanding

Excellent

(A+, A, A-) Exhibits a thorough grasp of time to biostatistics concepts and methods and effectively applies them to intricate problems

Good

(B+, B) Displays adequate understanding of biostatistics concepts and methods and effectively applies them to moderately challenging problems

Marginal

(B-, C+, C) Shows basic comprehension of biostatistics concepts and methods and applies them to standard problems

Failure

(F) Insufficient understanding of biostatistics concepts and methods and lacks the ability to apply them to problem-solving

Assessment Task

Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Problem solving based on comprehensive understanding

Excellent

(A+, A, A-) Consistently exhibits an in depth understanding of biostatistics concepts and methods and effectively applies them to real problems

Good

(B+, B) Effectively applies biostatistics concepts and methods to moderately challenging problems

Marginal

(B-, C+, C) Applies biostatistics concepts and methods to standard problems and demonstrates a basic understanding

Failure

(F) Lacks understanding of biostatistics concepts and methods and cannot apply them to problem-solving

Part III Other Information

Keyword Syllabus

Causal inference, genomic analysis, high-dimensional data analysis, post-model selection, reinforcement learning, survival analysis, time series analysis, regularised regression, generalized linear models, regression for counts

Reading List

Compulsory Readings

Title	
1	Nil

Additional Readings

Title	
1	Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction (Cambridge University Press; 1st edition), by Guido W. Imbens and Donald B. Rubin
2	High-Dimensional Statistics (Cambridge University Press), by Martin J. Wainwright
3	Foundations of Linear and Generalized Linear Models, by Alan Agresti
4	An Introduction to Statistical Learning with Applications in R, by Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani
5	Extending the Linear Model with R: Generalized Linear, Mixed Effects and Nonparametric Regression Models, by Julian J. Faraway
6	Regression Analysis of Count Data, by A. Colin Cameron and Pravin K. Trivedi